

Yuan-Yen Peng

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Research Interests

- LHC Higgs, electroweak, and diboson, BSM-sensitive measurements with Run 3 data
- W-boson polarization, quantum correlations, and spin-sensitive observables
- Physics-guided reconstruction and inference for partially observed final states
- Trigger-aware and detector-facing ML for future LHC measurements

Education

MSc in Physics, National Tsing Hua University GPA: 4.26 / 4.30 Thesis: Quantum Entanglement Measurements in the $H \rightarrow WW^* \rightarrow \ell\nu\ell\nu$ Decay Channel with the ATLAS Detector 	Sept. 2023 – June 2025 Hsinchu, Taiwan
BSc in Physics, National Tsing Hua University GPA: 3.98 / 4.30 (Last 60 credits: 4.12)	Aug. 2020 – June 2023 Hsinchu, Taiwan
Asia-Europe-Pacific School of High-Energy Physics CERN, KEK, CEA, CNRS, DESY	(forthcoming) Oct. 2026 Taoyuan, Taiwan
Experimental High Energy Physics Summer School Taiwan Instrumentation and Detector Consortium	June 2022 Taipei, Taiwan
Tsing Hua Interdisciplinary Program National Tsing Hua University	Sept. 2019 – July 2020 Hsinchu, Taiwan

Academic Experience

Research Assistant – R&D Substitute Service in NTHU	July 2025 – Jan. 2027
◦ Developed ML-based methods for W-boson kinematics in $H \rightarrow WW^* \rightarrow \ell\nu\ell\nu$, emphasizing spin correlations and integration into the analysis framework. ◦ Contributed to ATLAS internal note (ANA-HIGP-2024-27-INT1) on ML-based reconstruction and W-polarization interpretation. ◦ Contributed to the ATLAS Run 3 HWW coupling analysis. ◦ Co-developed the ATLAS HGTD production database for the detector upgrade. ◦ Built containerized ATLAS/ML environments with local CernVM-FS caching. ◦ Co-mentored MSc students on HWW analysis.	
Research Assistant – NTHU	Sept. 2022 – June 2025
◦ Studied quantum entanglement in $H \rightarrow WW^* \rightarrow \ell\nu\ell\nu$ with the ATLAS detector. ◦ Delivered ATLAS talks on HWW reconstruction and quantum entanglement studies.	
Research Student – NTHU	Feb. 2021 – Sept. 2022
◦ Early-stage ATLAS analysis experience and hands-on HEP software frameworks.	

Teaching Assistant

Dept. of Physics in NTHU

Feb. 2024 – Jan. 2025

Hsinchu, Taiwan

- Graduate-level Quantum Mechanics I
- Thermal and Statistical Physics I
- General Physics B II

Selected Presentations and ATLAS Activities

ATLAS NRML Workshop

at Nikhef

Apr. 2025

Amsterdam, Netherlands

- [[Oral presentation](#)]: *Reconstructions of W and ν with Machine Learning*

ATLAS Collaboration Week

at CERN

Oct. 2024

Geneva, Switzerland

- [[Oral presentation](#)]: *Reconstruction of W Boson Rest Frames in $H \rightarrow WW^* \rightarrow \ell\nu\ell\nu$*

Annual Meeting of the Physical Society of Taiwan

at Natl. Sun Yat-sen Univ.

Jan. 2025

Kaohsiung, Taiwan

- [[Poster presentation](#)]: *Quantum Entanglement in $H \rightarrow WW^* \rightarrow \ell\nu\ell\nu$*

ATLAS HWW Workshop

at Uni Freiburg

Mar. 2023

Freiburg, Germany

- [[Contributed talk](#)]: Presented in pre-workshop focus session.

HWW Quantum Entanglement Kick-off Meeting

via Virtual Meeting

Apr. 2024

Online

- NTHU-group contributor to the HWW quantum-entanglement study.

ATLAS Higgs Workshop

at UTokyo

Oct. 2022

Tokyo, Japan

- Contributed to the HWW quantum-entanglement presentation.

Honors and Scholarships

Phi Tau Phi Scholastic Honor Society Honorary Member

June 2025

College of Science Elite Student Award, NTHU

April 2025

Maxwell Scholarship, Dept. of Physics, NTHU (merit-based)

May 2024

Ms. Ru-Bao Huang Li Physics Memorial Scholarship, NTHU

Apr. 2024

Mr. Hong-Lin Wu Memorial Scholarship (academic merit)

June 2021

Outstanding Academic Achievement Award, NTHU ($\times 3$)

2020 – 2023

Technical Skills

ML: PyTorch/Lightning, FrEIA, ONNX, TensorFlow/Keras, W&B

HEP: ROOT, FastFrames, uproot, mplhep

Database & Web: Django, Vue.js, HTML

Programming: Python, C++

Scientific Computing: Git, Docker/Podman, CernVM-FS, UNIX

Markup tools: $\text{\LaTeX}$, Markdown, conventional office suites

Languages: English (TOEFL iBT 101), Mandarin Chinese (native)